2022 SSMO Speed Round Solutions

SMO Team

SR 1: Bobby is bored one day and flips a fair coin until it lands on tails. Bobby wins if the coin lands on heads a positive even number of times in the sequence of tosses. The probability that Bobby wins can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{7}$

Solution: Bobby wins if the sequence consists of $2k$ heads followed by a tail for some integer $k \geq 1$. The probability of this occurring is

$$\sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^{2k+1} = \frac{1/8}{1-1/4} = \frac{1}{6} \implies 1+6 = \boxed{7}.$$

SR 2: A bag is big enough to hold exactly 6 large pencils, 12 medium pencils, or 30 small pencils, with no space left over. Given that there is 1 large pencil and 3 medium pencils currently in the bag, what is the maximum number of small pencils that may be added to the bag?

Answer: $\boxed{17}$

Solution: Each large pencil takes up $\frac{1}{6}$ of the bag, each medium pencil takes up $\frac{1}{12}$ of the bag, and each small pencil takes up $\frac{1}{30}$ of the bag. Thus, the maximum number of small pencils is

$$\left\lfloor \frac{1-\frac{1}{6}-3\cdot\frac{1}{12}}{\frac{1}{30}} \right\rfloor = \left\lfloor \frac{\frac{7}{12}}{\frac{1}{30}} \right\rfloor = \lfloor 17.5 \rfloor = \boxed{17}.$$

SR 3: Let $ABCD$ be a parallelogram. Let E be a point on CD such that $\frac{CE}{DE} = \frac{2}{3}$. Suppose that BE and AC intersect at F and the area of triangle AEF is 15. Find the area of $ABCD$.

Answer: $\boxed{105}$

Solution: Note that $\triangle CEF \sim \triangle ABF$. We have

$$\begin{aligned}
[AEF] &= [AEC] - [CEF] \\
&= \frac{CE}{CD} \cdot [ACD] - \frac{CE}{CD} \cdot \frac{CE}{CE+AB} \cdot [ACD] \\
&= \frac{CE}{CE+DE} \cdot [ACD] - \frac{CE}{CE+DE} \cdot \frac{CE}{CE+CD} \cdot [ACD] \\
&= \frac{2}{2+3} \cdot [ACD] - \frac{2}{2+3} \cdot \frac{CE}{CE+(CE+DE)} \cdot [ACD] \\
&= \frac{2}{5} \cdot [ACD] - \frac{2}{5} \cdot \frac{2}{2+(2+3)} \cdot [ACD] \\
&= \left(\frac{2}{5} - \frac{2}{5} \cdot \frac{2}{7} \right) [ACD] \\
&= \frac{2}{7}[ACD] = \frac{2}{7} \cdot \frac{1}{2} \cdot [ABCD] = \frac{1}{7}[ABCD].
\end{aligned}$$

Thus,

$$[ABCD] = 7[AEF] = 7 \cdot 15 = \boxed{105}.$$

SR 4: Consider a quadrilateral $ABCD$ with area 120 and satisfying $AB + CD = AD + BC = 24$. There exists a point P in 3D space such that the distances from P to AB, BC, CD, and DA are all equal to 13. Find the volume of $PABCD$.

Answer: $\boxed{480}$

Solution: Let r be the inradius of $ABCD$ and h be the height of P. Note that

$$\tfrac{1}{2}r(AB + BC + CD + DA) = [ABCD] \implies r = \frac{2 \cdot 120}{24 + 24} = 5.$$

We have

$$h^2 + 5^2 = 13^2 \implies h = 12,$$

so the volume of $PABCD$ is

$$\tfrac{1}{3}h[ABCD] = \tfrac{1}{3} \cdot 12 \cdot 120 = \boxed{480}.$$

SR 5: Let $ABCD$ be a square such that E is on AD and F is on CD. If $AE = DF$ and $\frac{[BEF]}{[ABCD]} = \frac{7}{18}$, then the value of $\frac{EF^2}{BC^2}$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{14}$

Solution: WLOG assume $ABCD$ is a unit square and let $AE = DF = a$. Then

$$\frac{[BEF]}{[ABCD]} = \frac{7}{18} \implies [BEF] = \frac{7}{18}.$$

Note that

$$\begin{aligned}[BEF] &= [ABCD] - [AEB] - [DEF] - [BFC] \\ &= 1 - \frac{AE \cdot AB}{2} - \frac{DE \cdot DF}{2} - \frac{FC \cdot BC}{2} \\ &= 1 - \frac{a}{2} - \frac{(1-a)a}{2} - \frac{1-a}{2} \\ &= \frac{a^2 - a + 1}{2},\end{aligned}$$

so

$$\frac{a^2 - a + 1}{2} = \frac{7}{18} \implies a^2 - a = -\frac{2}{9}.$$

Now

$$\begin{aligned}EF^2 = DE^2 + DF^2 = (1-a)^2 + a^2 = 2a^2 - 2a + 1 \\ = 2(a^2 - a) + 1 = 2\left(-\frac{2}{9}\right) + 1 = \frac{5}{9}.\end{aligned}$$

Hence

$$\frac{EF^2}{BC^2} = EF^2 = \frac{5}{9} \implies 5 + 9 = \boxed{14}.$$

SR 6: At the beginning of day 1, there is a single bacterium in a petri dish. During each day, each bacterium in the petri dish divides into $a > 1$ new bacteria, and $b \geq 1$ bacteria are added to the petri dish. For example, at the end of day 1, there are $a + b$ bacteria in the petri dish. If, at the end of day 4, the number of bacteria in the petri dish is a multiple of 48, find the minimum possible value of $a + b$.

Answer: $\boxed{36}$

Solution: We have

$$\begin{aligned}a + b &\quad \text{bacteria at the end of day 1,} \\ a^2 + ab + b &\quad \text{bacteria at the end of day 2,} \\ a^3 + a^2b + ab + b &\quad \text{bacteria at the end of day 3, and} \\ a^4 + a^3b + a^2b + ab + b &\quad \text{bacteria at the end of day 4.}\end{aligned}$$

So,

$$a^4 + a^3b + a^2b + ab + b \equiv 0 \pmod{48}.$$

We factor 48 as $2^4 \cdot 3$. We must have

$$a^4 + a^3b + a^2b + ab + b \equiv 0 \pmod{16}.$$

If $a \equiv 1 \pmod 2$, then $4b + 1 \equiv 0 \pmod 2$, which is impossible. So, we must have $a \equiv 0 \pmod 2$. Since $a^4 \equiv 0 \pmod{16}$ and $a^3 + a^2 + a + 1 = (a+1)(a^2+1)$ is odd, this implies $b \equiv 0 \pmod{16}$.

Now, from

$$a^4 + a^3b + a^2b + ab + b \equiv 0 \pmod 3,$$

we have

$$a \equiv 0 \pmod 3 \implies b \equiv 0 \pmod 3,$$

$$a \equiv 1 \pmod 3 \implies 1 + 4b \equiv 0 \pmod 3 \implies b \equiv 2 \pmod 3, \text{ or}$$
$$a \equiv 2 \pmod 3 \implies 16 + 15b \equiv 0 \pmod 3,$$

which is impossible. If $a \equiv 0 \pmod 3$, then we have

$$a \equiv 0 \pmod 6 \quad \text{and} \quad b \equiv 0 \pmod{48}.$$

If $a \equiv 1 \pmod 3$, then we have

$$a \equiv 4 \pmod 6 \quad \text{and} \quad b \equiv 32 \pmod{48}.$$

The first case gives $a + b = 6 + 48 = 54$ while the second case gives $a + b = 4 + 32 = \boxed{36}$.

SR 7: Let $A_1 = (1, 0)$. Define A_{n+1} as the image of A_n under a rotation of either $45°$, $90°$, or $135°$ clockwise about the origin, with each choice having a $\frac{1}{3}$ chance of being selected. Find the expected value of the smallest positive integer $n > 1$ such that A_n coincides with A_1.

Answer: $\boxed{9}$

Solution: Since each rotation is a multiple of $45°$, let A_n be the k_nth of the 8 points $e^{-k\pi i/4}$, so each step adds 1, 2, or 3 to k_n modulo 8 with equal probability. Since each residue is reached from exactly three residues, each with probability $\frac{1}{3}$, the uniform distribution on the 8 residues is stationary. So, from Kac's lemma, the expected number of steps to first return to $k = 0$ is 8, meaning the answer is $1 + 8 = \boxed{9}$.

SR 8: How many positive integers cannot be written as $7a + 19b + 28c$, where a, b, and c are positive integers (not necessarily distinct)?

Answer: $\boxed{107}$

Solution: We rewrite

$$7a + 19b + 28c = 7(a + 4c) + 19b \geq 7 \cdot 5 + 19 = 54.$$

So, all $N \leq 53$ cannot be written in the form. Since $\gcd(7, 19) = 1$, there are

$$\frac{(7 - 1)(19 - 1)}{2} = 54$$

positive integers that cannot be written as $7x + 19y$, for $x, y \geq 0$. Since $a + 4c$ takes every integer ≥ 5 and $b - 1 \geq 0$, for $N \geq 54$, N cannot be written as $7a + 19b + 28c$ if and only if $N - 54 = 7(a + 4c - 5) + 19(b - 1)$ is one of the 54 numbers above. So, we have an answer of

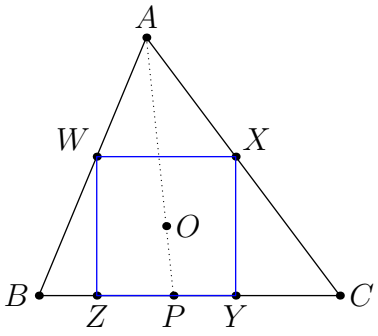
$$53 + 54 = \boxed{107}.$$

SR 9: Consider a triangle ABC such that $AB = 13$, $BC = 14$, $CA = 15$ and a square $WXYZ$ such that Y and Z lie on $\overleftrightarrow{BC}$, W lies on $\overleftrightarrow{AB}$, and X lies on $\overleftrightarrow{CA}$. Suppose further that W, X, Y, and Z are distinct from A, B, and C. Let O be the center of $WXYZ$. If AO intersects BC at P, then the sum of all values of $\frac{BP}{CP}$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{29}$

Solution: We will use analytic geometry. Let $A = (5, 12)$, $B = (0, 0)$, and $C = (14, 0)$. We have

$$\overleftrightarrow{BC} : y = 0, \quad \overleftrightarrow{AB} : y = \frac{12}{5}x, \text{ and } \quad \overleftrightarrow{CA} : y = -\frac{4}{3}x + \frac{56}{3}.$$



First, assume that $WXYZ$ is completely contained within the triangle. Let $O = (a, b)$. We have $W = (a - b, 2b)$ and $X = (a + b, 2b)$. Substituting into $\overleftrightarrow{AB}$ and $\overleftrightarrow{AC}$, we have

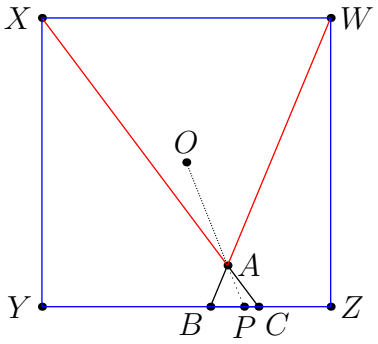
$$2b = \frac{12}{5}(a - b) \quad \text{and} \quad 2b = -\frac{4}{3}(a + b) + \frac{56}{3}.$$

Solving, we get

$$(a, b) = \left(\frac{77}{13}, \frac{42}{13}\right) \implies P = \left(\frac{119}{19}, 0\right),$$

giving

$$\frac{BP}{CP} = \frac{\frac{119}{19}}{14 - \frac{119}{19}} = \frac{17}{21}.$$



Next, assume $WXYZ$ is outside the triangle. Let $O = (a, b)$. We have $W = (a + b, 2b)$ and $X = (a - b, 2b)$. Substituting into $\overleftrightarrow{AB}$ and $\overleftrightarrow{AC}$, we have

$$2b = \frac{12}{5}(a + b) \quad \text{and} \quad 2b = -\frac{4}{3}(a - b) + \frac{56}{3}.$$

Solving, we have

$$(a, b) = (-7, 42) \implies P = \left(\frac{49}{5}, 0\right),$$

giving

$$\frac{BP}{CP} = \frac{\frac{49}{5}}{14 - \frac{49}{5}} = \frac{7}{3}.$$

Our answer is thus

$$\frac{17}{21} + \frac{7}{3} = \frac{22}{7} \implies 22 + 7 = \boxed{29}.$$

SR 10: Let $S = \{2, 7, 15, 26, \ldots\}$ be the set of all numbers for which the ith element in S is the sum of the ith triangular number and the ith positive perfect square. Let T be the set which contains all unique remainders when the elements in S are divided by 2022. Find the number of elements in T.

Answer: $\boxed{1014}$

Solution: Note that $2022 = 2 \cdot 3 \cdot 337$. The ith term is

$$S_i = \frac{i(i+1)}{2} + i^2 = \frac{3i^2 + i}{2}.$$

We analyze the possible residues of S_i modulo each prime factor of 2022. From direct checking, S_i attains all residues modulo 2 and modulo 3. Since 337 is prime, we work modulo 337 and consider $3i^2 + i \pmod{337}$. Suppose there exist distinct residues a and b such that

$$\frac{3a^2 + a}{2} \equiv \frac{3b^2 + b}{2} \pmod{337},$$

which implies

$$\frac{(3a^2 + a) - (3b^2 + b)}{2} \equiv 0 \pmod{337}.$$

Simplifying gives

$$(3a + 3b + 1)(a - b) \equiv 0 \pmod{337},$$

so since $a \neq b$, we must have $3a + 3b + 1 \equiv 0 \pmod{337}$, or $a + b \equiv 112 \pmod{337}$. Thus, for each residue $a \not\equiv 56 \pmod{337}$, there is a unique distinct residue b mapping to the same value, and $a \equiv 56$ maps to itself. Therefore, the number of distinct residues modulo 337 is $\frac{337+1}{2} = 169$.

Since S_i modulo 2, 3, and 337 depends only on i modulo 4, 3, and 337, respectively, and these moduli are pairwise relatively prime, every combination of residues is attained, from the Chinese Remainder Theorem. So, the number of distinct remainders modulo 2022 is

$$2 \cdot 3 \cdot 169 = \boxed{1014}.$$